Ghosts, Geometry, and Reality in Fourth-Order Quantum Theories

A guided introduction to the Pais–Uhlenbeck oscillator, Krein quantization, and higher-derivative gravity

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Abstract

Higher-derivative theories appear throughout mathematical physics and quantum gravity. Their equations contain more time derivatives than ordinary Newtonian or quantum systems, which gives them additional solutions and often improves their ultraviolet behavior. The price is the appearance of so-called ghosts: modes associated with negative energy, negative norm, or an indefinite state-space geometry. It is often said that such theories are therefore inconsistent. That conclusion is too quick: "the ghost problem" is not one problem but several — classical stability, spectral boundedness, positivity of probabilities, the choice of real observables, the construction of the vacuum, the representation of the field algebra, and the behavior of the theory when distinct modes merge into a Jordan block. Earlier work by Bender and Mannheim showed that the unequal-frequency Pais–Uhlenbeck oscillator, the simplest fourth-order system, can be converted into two positive oscillators by a nonunitary similarity transformation and quantized with a positive inner product; more recently, Bateman and Turok developed a different treatment of the resonant massless theory using a Krein space and a hidden ghost-parity symmetry. The five technical papers introduced here do not replace those constructions. They explain their mathematical status, classify their ambiguities, identify the geometry that selects the Bender–Mannheim metric, connect the positive and Krein descriptions through a common complex vacuum covariance, and determine what changes when gauge symmetry and Lorentz covariance are imposed in fourth-order gravity. The resulting picture is that a fourth-order theory may possess one complex spectral dynamics but several inequivalent real forms and completions: the positive-metric and Krein descriptions can agree on complex correlation functions while disagreeing about which fields are real, which operators are Hermitian, and how probabilities are represented. At resonance the positive diagonalizing metric terminates, but the vacuum covariance can remain regular and continue into a Jordan theory. Extending the free classification to interactions, a fifth paper formulates a deformation problem for the positive metric and the Krein involution. The positive Pais–Uhlenbeck metric deforms perturbatively at generic frequencies, but exact energy-conserving conversion processes generate obstruction classes. In the oscillator these occur on isolated rational resonance loci and can produce complex-conjugate spectral pairs; in the field theory, continuous momentum makes the corresponding branch-changing scattering

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shells generic. The massless perfect-square theory nevertheless possesses an exact nonlinear two-sector involution, showing that failure of a positive pointed-sector completion does not imply absence of a doubled Krein-type structure.

Contents

1 Why higher derivatives create ghosts

Most familiar equations of motion contain at most two time derivatives. Newton's equation has the form

$$m\ddot{q} = F(q), \tag{1}$$

and the Klein–Gordon field equation has the form

$$(\Box + m^2)\phi = 0, \tag{2}$$

where the empty square is the standard symbol for the wave operator (d'Alembertian), $\Box = \partial^2/\partial t^2 - \nabla^2$ — the spacetime analogue of the Laplacian. Specifying a solution requires two pieces of initial data: position and velocity for a particle, or the field and its first time derivative for a field theory.

A fourth-order equation such as

$$\left(\frac{d^2}{dt^2} + \omega_1^2\right)\left(\frac{d^2}{dt^2} + \omega_2^2\right) z(t) = 0 \tag{3}$$

requires four pieces of initial data. Its solution contains two independent frequencies,

$$z(t) = A_1 e^{-i\omega_1 t} + B_1 e^{i\omega_1 t} + A_2 e^{-i\omega_2 t} + B_2 e^{i\omega_2 t}. \tag{4}$$

The extra derivatives have introduced an extra oscillator.

That sounds harmless. The difficulty is that the standard Hamiltonian formulation usually assigns opposite signs to the two oscillators:

$$H \;\sim\; H_{\omega_1} - H_{\omega_2}. \tag{5}$$

One branch then appears to carry negative energy. If one changes the quantization convention to keep the energy positive, the same branch may instead acquire negative norm. This is the traditional higher-derivative ghost problem.

The classical theorem of Ostrogradsky [1] explains why the Hamiltonian obtained from a nondegenerate higher-derivative Lagrangian is generically unbounded in its canonical momenta (see [2] for a modern review). But that theorem does not by itself answer every quantum question. The questions divide into two stages.

Stage 1 — the free theory. Can the Hamiltonian be diagonalized? Can a positive inner product be found? Which variables count as real? What happens at the Jordan limit where two frequencies coincide?

Stage 2 — interaction stability. Can the inner product be deformed when interactions are added? Can ghost parity remain conserved when particles convert into one another?

Do energy-conserving processes obstruct the deformation? Does an exact doubled symmetry survive even if each individual sector fails?

Solving the free Hamiltonian does not settle the interacting ghost problem. A free ghost resolution is analogous to balancing a machine while its parts move independently; interactions test whether the balance survives when the parts exchange energy. Both stages are central to everything that follows: Sections 1–13 treat the free theory, Section 14 the interacting one, and Sections 15–17 the gravitational application.

2 The Pais–Uhlenbeck oscillator as a microscope

The Pais–Uhlenbeck (PU) oscillator [3] is the simplest model containing the full fourth-order difficulty. Its Lagrangian can be written

$$L = \frac{\gamma}{2}\left(\ddot{z}^{\,2} - (\omega_1^2+\omega_2^2)\,\dot{z}^{\,2} + \omega_1^2\omega_2^2\,z^2\right), \qquad \omega_1 > \omega_2 > 0. \tag{6}$$

Its equation of motion factorizes:

$$\left(\frac{d^2}{dt^2}+\omega_1^2\right)\left(\frac{d^2}{dt^2}+\omega_2^2\right)z = 0. \tag{7}$$

It therefore contains two ordinary frequencies, but their canonical combination has the wrong-sign structure characteristic of a ghost.

The oscillator is useful because it is finite dimensional and exactly solvable. Questions that become difficult analytic problems in a field theory reduce here to matrix algebra:

- Can the Hamiltonian be transformed to a positive normal form?
- Is that transformation unique?
- How many positive metrics exist?
- Is there a preferred one?
- What happens as $\omega_1 \to \omega_2$?
- Does the limiting Jordan system admit any positive invariant inner product?

A correct answer at the oscillator level does not automatically solve a quantum field theory. But the oscillator provides the local building block for each momentum mode of a free fourth-order field.

3 What Bender and Mannheim had already solved

Bender and Mannheim [5, 6] observed that the PU Hamiltonian should not be quantized using the naive real contour and inner product. After a complex continuation of one canonical coordinate, the Hamiltonian becomes non-Hermitian but PT-symmetric [4].

They constructed an operator

$$Q = \alpha\,pq + \beta\,xy \tag{8}$$

and the associated similarity transformation $\rho = e^{-Q/2}$ such that

$$h = \rho \, H_{\mathrm{PT}} \, \rho^{-1} \tag{9}$$

is a conventional positive Hamiltonian describing two ordinary oscillators. The original Hamiltonian is then Hermitian not with respect to the naive adjoint, but with respect to the positive metric

$$\eta = \rho^{\dagger}\rho = e^{-Q}, \tag{10}$$

in the general framework of pseudo-Hermitian quantum mechanics [8, 9]. The physical inner product becomes

$$\langle \psi, \phi \rangle_{\eta} = \langle \psi, \eta \, \phi \rangle. \tag{11}$$

With this inner product, the unequal-frequency oscillator has positive norms and a positive spectrum. Thus an important version of the ghost problem was already solved:

The free unequal-frequency Pais–Uhlenbeck oscillator admits a positive-metric quantization.

Three levels must nevertheless be distinguished, and the series addresses each in turn: the *free spectral solution* (the free oscillator maps to a positive two-oscillator Hamiltonian); *perturbative interaction stability* (the metric can be deformed away from on-shell resonances); and *global interacting viability* (energy-conserving conversion processes may obstruct the deformation and, in finite resonant systems, produce complex spectra). In short:

The free ghost problem was solved locally; Paper V determines where that solution survives interaction.

What remained unclear was the mathematical status of the construction. Why does the particular operator Q appear? Is it an ansatz or a canonical object? Is the metric unique? If it is not unique, why should the Bender–Mannheim metric be preferred? What happens when the oscillator becomes a field with infinitely many momentum modes? How does this positive construction relate to an indefinite Krein-space construction [7]? And what survives after gauge reduction in gravity? The four papers [12, 13, 14, 15] address these questions.

4 Paper I: reconstructing the Bender–Mannheim transformation

The first paper [12] reformulates the oscillator as a problem in complex symplectic geometry. Collect the canonical variables into a vector $V = (x, y, p, q)^{\top}$, with commutation relations encoded by the symplectic matrix J:

$$[V_a, V_b] = i J_{ab}. \tag{12}$$

A quadratic Hamiltonian has the form $H = \frac{1}{2} V^{\top} G V$, where G is a symmetric matrix, and the Hamiltonian flow is generated by $A = JG$. The problem is to find a complex symplectic matrix S satisfying

$$S^{\top} J S = J, \qquad S^{\top} G S = G_0, \tag{13}$$

where G_0 is the positive two-oscillator normal form.

The first result is that, after a specified canonical normalization, this problem has a *unique* Hermitian positive-definite solution S_+. The complete solution family is the coset

$$S_+H, \qquad H \cong SO(2,\mathbb{C}) \times SO(2,\mathbb{C}), \tag{14}$$

where H is the stabilizer of the positive normal form. Thus many complex symplectic transformations diagonalize the Hamiltonian, but exactly one normalized representative is both Hermitian and positive.

The logarithm of this matrix is then converted into a quadratic quantum operator through the complexified metaplectic representation [17]. The resulting operator is precisely $Q = \alpha pq + \beta xy$ with the coefficients found by Bender and Mannheim. The logical direction has therefore been reversed:

$$\boxed{(G,\, J,\, G_0) \;\longrightarrow\; S_+ \;\longrightarrow\; Q.}$$

The operator Q is no longer introduced by guessing a useful quadratic expression. It is reconstructed from the symplectic normal-form problem.

4.1 The exact spectrum of Q

In normalized oscillator coordinates,

$$Q = r\left(a_1 a_2^\dagger + a_1^\dagger a_2\right), \qquad r = \log\frac{\omega_1+\omega_2}{\omega_1-\omega_2}. \tag{15}$$

This is a number-preserving Schwinger $SU(2)$ generator. It commutes with $N_{\text{tot}} = N_1 + N_2$, and on the fixed-$N$ subspace its eigenvalues are $r(-N, -N+2, \ldots, N-2, N)$. Consequently

$$\operatorname{spec} Q = r\,\mathbb{Z}, \tag{16}$$

with pure-point spectrum and infinite multiplicities. The metric has spectrum

$$\operatorname{spec}(e^{-Q}) = \{e^{-rn} : n \in \mathbb{Z}\} \cup \{0\}, \tag{17}$$

where zero is an accumulation point rather than an eigenvalue. This provides a natural invariant domain: the algebraic Fock space of finite linear combinations of number states, preserved by Q, $e^{\pm Q/2}$, and the quadratic Hamiltonians appearing in the construction.

4.2 Many metrics, one canonical Gaussian metric

Pseudo-Hermiticity alone does not make the metric unique. If $h = \rho H_{\text{PT}} \rho^{-1}$ and W is a positive operator commuting with h, then

$$\eta_W = \rho^\dagger W \rho \tag{18}$$

also defines an algebraic positive metric form [8, 9]. Thus the complete metric freedom is larger than the finite-dimensional symplectic stabilizer: the stabilizer describes the Gaussian or quadratic part of the ambiguity, while arbitrary positive functions of the normal-mode number operators provide additional non-Gaussian freedom. The first paper therefore distinguishes the *unique normalized positive symplectic diagonalizer* from the *nonunique pseudo-Hermitian metric operator.*

5 Paper II: why the canonical metric is the least-distorting one

Uniqueness inside a restricted class does not by itself explain why S_+ is geometrically natural. The second paper [13] supplies a variational characterization. For an admissible diagonalizer S, define

$$\mathcal{F}(S) = \left\| \log(S^\dagger S) \right\|_F^2. \tag{19}$$

The positive matrix $S^\dagger S$ measures how much the transformation stretches and contracts the canonical coordinates; the logarithm converts multiplicative stretching into additive "rapidities," and the Frobenius norm measures their total size. The theorem is

$$\mathcal{F}(S) \;\geq\; \mathcal{F}(S_+) = 4r^2, \tag{20}$$

with equality precisely for S_+ multiplied by a unitary element of the stabilizer — which for the PU oscillator consists of real sign matrices, so all minimizers have the same positive polar factor and determine the same Bender–Mannheim metric. In ordinary language:

S_+ is the least-distorting allowed transformation that makes the Hamiltonian positive.

This is analogous to finding the shortest path from a point to a surface: a shortest path meets a smooth surface orthogonally. Here the "surface" is the orbit generated by transformations that preserve the positive normal form.

5.1 Cartan projection

The geometry is that of the complex symplectic group $\mathrm{Sp}(4,\mathbb{C})$ and its maximal compact subgroup $K = \mathrm{Sp}(4,\mathbb{C}) \cap U(4)$, acting on the space of positive matrices with the affine-invariant metric [18]. The Cartan projection records the logarithmic singular values of a transformation. With the doubled Gram convention used in the papers ($\mu(S) = \log \operatorname{spec}(S^\dagger S)$),

$$\mu(S_+) = (r, r), \tag{21}$$

a point on the wall $x_1 = x_2$ of the C_2 Weyl chamber. The distortion functional is related to the Cartan norm by

$$\mathcal{F}(S) = 2\|\mu(S)\|^2. \tag{22}$$

Thus the minimum-distortion theorem is a Cartan-projection theorem.

5.2 Why the theorem is not a routine Kempf–Ness application

The physical stabilizer $H \cong SO(2,\mathbb{C})^2$ is not compatible with the canonical Cartan involution determined by the physical inner product: generically

$$H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \tag{23}$$

rather than the maximal compact $U(1)^2$ of the abstract group. The obstruction belongs to the *embedding* of H, not to its abstract group structure, so the classical nearest-point theory of self-adjoint groups [19, 20] does not apply to H directly.

The resolution is to enlarge H to its smallest compatible (θ-stable) hull. Generically this is

$$SL(2,\mathbb{C}) \times SL(2,\mathbb{C}), \tag{24}$$

the mode-preserving subgroup, whose positive part is a totally geodesic copy of $\mathbb{H}^3 \times \mathbb{H}^3$. The Bender–Mannheim direction is orthogonal to this mode-preserving geometry, so standard nonpositively curved projection theory can be applied to the hull and the result restricted back to the physical stabilizer. The paper proves a general principle:

> *A positive diagonalizer minimizes Cartan distortion along a compatible subgroup exactly when its Hermitian generator is orthogonal to the subgroup's positive tangent directions.*

For a splitting into many oscillator modes, those tangent directions are the mode-preserving deformations, and the normal directions are precisely the inter-mode couplings. The PU oscillator is the simplest nontrivial example: its preferred generator is entirely inter-mode.

6 The exceptional point: when two frequencies merge

The rapidity is $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$. As $\omega_1 \to \omega_2$, the denominator tends to zero and $r \to \infty$. The positive diagonalizer stretches one pair of directions by $e^{r/2}$ and contracts another by $e^{-r/2}$; the corresponding Cartan distance diverges.

This is not merely a poor choice of coordinates. At equal frequency, the dynamical matrix ceases to be diagonalizable and develops a Jordan block — the hallmark of an exceptional point [22, 23]. A 2×2 Jordan block has the form

$$H_J = \begin{pmatrix} \omega & 1 \\ 0 & \omega \end{pmatrix}. \tag{25}$$

Suppose there were a positive-definite Hermitian form η satisfying $H_J^\dagger \eta = \eta H_J$. Then the matrix $\eta^{1/2} H_J \eta^{-1/2}$ would be Hermitian, hence diagonalizable. But similarity transformations preserve Jordan form. This is impossible. Hence a nontrivial Jordan block admits no nondegenerate positive-definite invariant Hermitian form [12, 21]. It may still admit:

- degenerate positive-semidefinite forms;
- nondegenerate indefinite forms;
- or intrinsically Jordan PT-symmetric realizations [6].

What fails is the continuation of the unequal-frequency positive Hilbert metric as a nondegenerate invariant positive form. This distinction matters: the conclusion is not that the equal-frequency theory cannot exist, but that it cannot remain an ordinary diagonalizable positive-metric system of the same kind.

7 From one oscillator to a field

Consider the free fourth-order scalar field

$$(\Box + m_1^2)(\Box + m_2^2)\,\phi = 0. \tag{26}$$

After Fourier transforming in space, each spatial momentum k contributes a PU oscillator with frequencies $\omega_i(k) = \sqrt{k^2 + m_i^2}$. The modewise rapidity is

$$r(k) = \log \frac{\big(\omega_1(k) + \omega_2(k)\big)^2}{m_1^2 - m_2^2} = 2\log|k| + O(1) \qquad (|k| \to \infty). \tag{27}$$

Thus the modewise similarity transformations become increasingly nonuniform in the ultraviolet.

This is where finite-dimensional and field-theoretic questions diverge. Each individual momentum mode can be treated successfully, while the collection of infinitely many modes may define an unbounded field transformation, a non-equivalent norm on phase space, a different quasifree representation, or a different real form of the field algebra. A modewise solution is therefore necessary but not sufficient for a field theory.

8 Metric geometry and vacuum geometry are different

A central lesson of the field papers [13, 14] is

$$\boxed{\text{metric distortion} \neq \text{vacuum excitation.}}$$

The Cartan rapidity $r(k)$ diverges like $2\log k$, and the finite-dimensional observable-space metric has eigenvalues $e^{\pm r(k)} \sim k^{\pm 2}$. This describes a strong deformation of the canonical phase-space norm. It does *not* follow that the transformation creates $O(k^2)$ particles.

In normalized coordinates the generator is number preserving:

$$Q_k = r(k)\big(a_{1,k}a_{2,k}^\dagger + a_{1,k}^\dagger a_{2,k}\big). \tag{28}$$

It acts inside the complexified beam-splitter algebra, and the canonical vacuum is annihilated by it, even though the associated Cartan norm is arbitrarily large.

The reason is geometric. Positive metrics are points in one quotient space, roughly G/K, while Gaussian vacuum rays belong to a different quotient, G/P_Ω, where P_Ω stabilizes the vacuum line. The vacuum-ray map does not factor through the metric quotient. Two transformations can therefore have very different metric distortion while producing the same vacuum ray. This prevents one from estimating physical particle content directly from the Cartan norm.

9 The universal complex vacuum covariance

For the fourth-order field, the selected two-point function has the spectral form

$$\widetilde{W}^+_{12}(p) = \varepsilon\,\theta(p^0)\,\frac{\delta(p^2 - m_1^2) - \delta(p^2 - m_2^2)}{m_1^2 - m_2^2}, \tag{29}$$

where $\varepsilon = \pm 1$ records the overall action convention. This is a divided difference of two ordinary Klein–Gordon two-point functions:

$$W_{12}^{+} = \varepsilon \, \frac{W_{m_1}^{+} - W_{m_2}^{+}}{m_1^2 - m_2^2}. \tag{30}$$

The field paper [14] shows that the admissible positive metrics all select the same complex covariance. They may change the physical adjoint and the representation of observables, but not this underlying spectral two-point distribution. The result should be stated carefully:

The complex covariance is metric-independent; the involution and completion need not be.

A state in the strict algebraic sense requires a fixed involution and a positivity condition. When different completions use different involutions, it is safer to speak first of the common complex quasifree covariance.

10 Resonance: the metric diverges but the covariance survives

Let $m_1^2, m_2^2 \to m^2$. Then the divided difference has the regular limit

$$\widetilde{W}_{mm}^{+}(p) = -\varepsilon \, \theta(p^0) \, \delta'(p^2 - m^2). \tag{31}$$

In the time domain, per mode,

$$W_{mm}^{+}(t, k) = -\varepsilon \, \frac{(1 + i\omega t) \, e^{-i\omega t}}{4\omega^3}, \qquad \omega = \sqrt{k^2 + m^2}. \tag{32}$$

The positive diagonalizing transformation diverges because the dynamics becomes Jordan. Yet the complex vacuum covariance remains finite. This is one of the main conceptual results:

Resonance is a singularity of the metric and dynamics, not necessarily of the vacuum covariance.

At the doubly massless point $m = 0$, the covariance is related, after matching the action sign and infrared convention, to the massless $\Box^2$ covariance used in the Krein-space construction of Bateman and Turok [7]. Their sign convention gives the opposite overall covariance to one common PU convention: the relation is an action-sign reversal, not a literal equality before conventions are matched.

11 What is a Krein space?

A Hilbert space has a positive inner product: $\langle \psi, \psi \rangle > 0$ for every nonzero state. A Krein space has a nondegenerate but *indefinite* inner product: a state can have positive, negative, or zero norm.

At first sight this appears incompatible with probability. But indefinite-metric quantum theories can possess an additional grading or symmetry that identifies the physical probability structure. In the Bateman–Turok construction [7], a hidden ghost-parity symmetry plays this role.

The positive and Krein descriptions sacrifice different conventional requirements. For an unpaired ghost oscillator, one may try to demand:

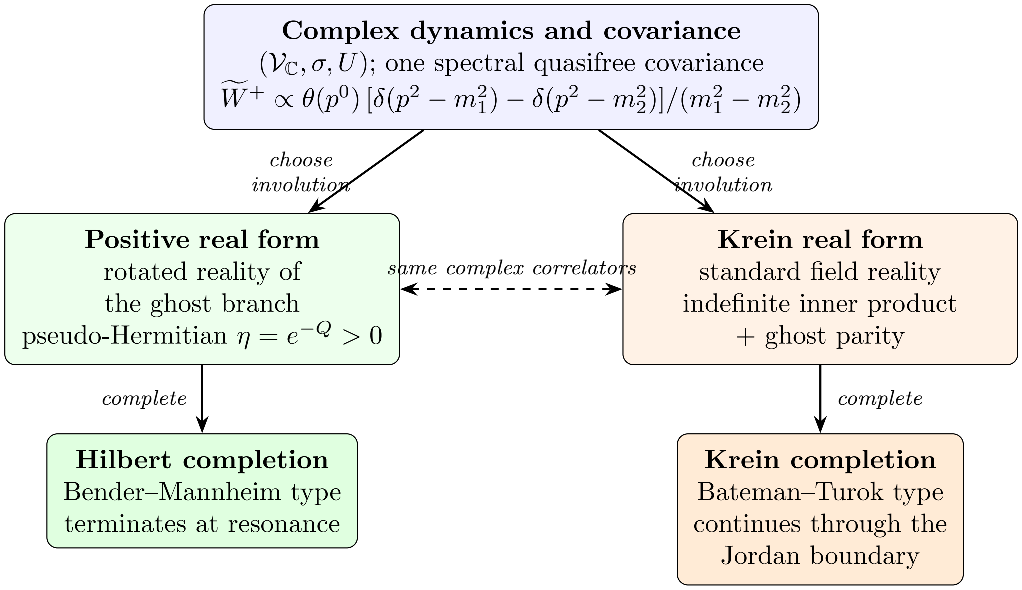

Figure 1: The three-level structure: one complex spectral covariance, two covariant real forms (involutions), two completions. The two branches agree on all complex correlation functions and differ in which operators are Hermitian and how probabilities are represented. At the resonant (Jordan) boundary only the Krein branch continues as a nondegenerate completion.

1. positive norm;
2. positive energy;
3. standard reality of the original field.

The three cannot all be retained simultaneously [15]. A complex quarter-turn of the ghost variables can make the energy and norm positive, but the original amplitude becomes anti-Hermitian; i times the amplitude is the physical Hermitian observable. Alternatively, the original field can remain real and the energy spectral condition can be retained, but the state space becomes indefinite. This gives two real forms of a common complex system:

	positive completion	Krein completion
energy	positive	positive
norm	positive	indefinite
original field reality	rotated	standard

The complex equations and complex covariance may agree even though the physical adjoints differ. Figure 1 summarizes the resulting three-level structure.

12 A necessary representation-theoretic distinction

There are two different ways to compare the positive PT construction with an ordinary oscillator Fock space, and conflating them produces contradictions [13].

12.1 Physical Dyson transport

For each mode, the PT ground state is exactly the Dyson pullback of the positive normal-form vacuum,

$$\psi_k = \rho_k^{-1}\phi_k. \tag{33}$$

Equip the PT space with the metric $\eta_k = \rho_k^\dagger \rho_k$. Then

$$\rho_k : \ \mathcal{H}_{\mathrm{PT},k} \longrightarrow \mathcal{H}_{\mathrm{normal},k} \tag{34}$$

is unitary with respect to the physical inner products, and $\rho_k \psi_k = \phi_k$ *exactly*. The pointed modewise unitaries therefore tensor together without any vacuum-overlap obstruction: the completed positive PT theory is globally equivalent to its positive normal-form theory.

12.2 Identity embedding in an auxiliary standard CCR algebra

One may instead place both Gaussian wavefunctions in the same standard Schrödinger L^2 space, retain the same canonical variables and standard adjoint, and compare the vectors through the identity embedding. Under that comparison, the per-mode Gaussian fidelity approaches $\sqrt{3}/2$, the expectation of the standard auxiliary number operator approaches $1/3$, and the infinite products of these auxiliary standard-CCR states are disjoint [24, 13].

This does not contradict the physical Dyson equivalence: the two comparisons identify observables differently.

$$\boxed{\text{identity embedding of auxiliary variables} \neq \text{Dyson transport of the physical algebra.}}$$

This is an example of why field representation theory cannot be discussed without specifying the algebra, its involution, the embedding between descriptions, and the chosen completion.

13 The fourth-order vacuum is locally well behaved

Ordinary four-dimensional Klein–Gordon two-point functions have a leading short-distance singularity proportional to $1/\sigma_+$, where $\sigma_+(x, y)$ is the regularized squared geodesic separation. In the fourth-order divided difference, this mass-independent leading term cancels. The result has the Hadamard-like form

$$W_{12}^+(x, y) = V_{12}(x, y)\, \log \sigma_+(x, y) + H_{12}(x, y), \tag{35}$$

where V_{12} and H_{12} are smooth and

$$V_{12}(x, x) = \frac{\varepsilon}{16\pi^2}. \tag{36}$$

There are further terms such as $\sigma \log \sigma$, so the two-point function is not simply a constant logarithm plus a smooth remainder. Nevertheless, its wavefront directions are the standard positive-frequency Hadamard directions, in the microlocal sense of [25].

This is the natural fourth-order analogue of the Hadamard property [14]. It says that the global ghost or completion problem does not automatically produce a pathological local ultraviolet singularity: the vacuum can be globally unusual in its real form or representation, yet locally well behaved in the microlocal sense required for renormalized observables [26, 27].

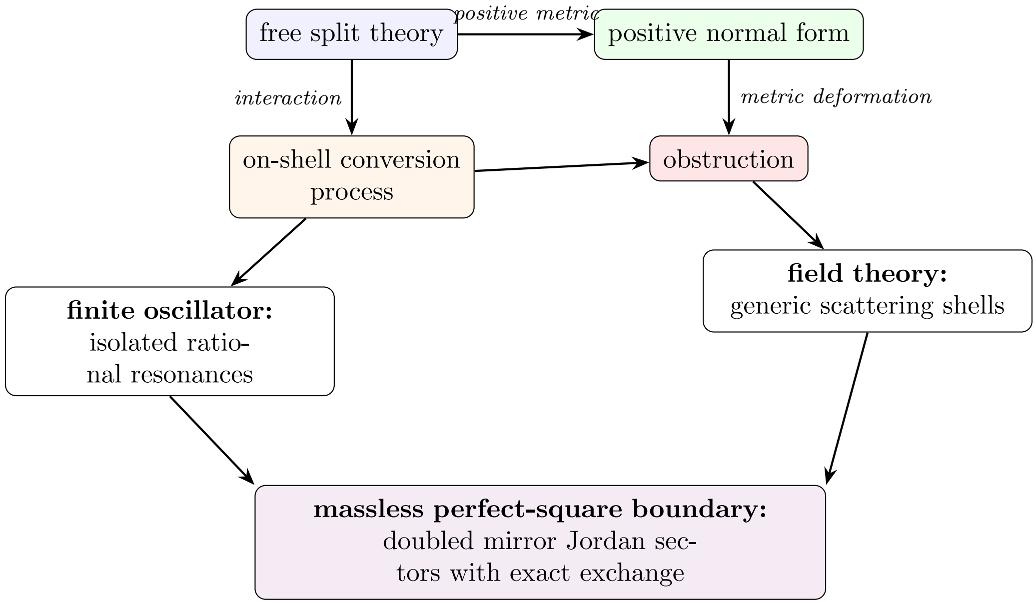

Figure 2: The interaction mechanism of [16]: metric deformation meets on-shell conversion; isolated resonances in the oscillator become generic scattering shells in the field theory; at the massless boundary a doubled exchange structure survives.

14 When interactions are switched on

Everything so far concerns free theories. The fifth paper of the series [16] asks the next question: can the free positive or Krein structure be continuously deformed when nonlinear processes are switched on? The answer has four parts: generically yes at low perturbative order away from resonances; no on specific energy-conserving conversion channels; in field theory such channels become generic because momentum is continuous; and an exact doubled sector-exchange symmetry can survive even though positive pointed-sector metrics fail. Figure 2 summarizes the mechanism.

14.1 Deforming the positive metric

Perturb the free theory, $H_\zeta = H_0+\zeta V$, and recall the free Dyson map with $\rho_0 H_0 \rho_0^{-1} = h_0 = h_0^\dagger$. One seeks a corrected map

$$\rho_\zeta = e^{-\frac{1}{2}(\zeta R_1+\zeta^2 R_2+\cdots)}\rho_0, \tag{37}$$

keeping the transformed Hamiltonian Hermitian order by order. At first order,

$$[h_0, R_1] = v^\dagger - v, \qquad v = \rho_0 V \rho_0^{-1}. \tag{38}$$

The interaction introduces a new “bad” non-Hermitian part, and the correction R_1 asks whether the measuring geometry can be adjusted to absorb it. The key principle is simple: a correction can remove terms that move states between *different* energies; it cannot remove a term that acts entirely among states with exactly the *same* energy. When two states have different energies, the metric can distinguish them; when they have exactly equal energy, the interaction is resonant and cannot be rotated away. Technically, the obstruction is the energy-conserving part of the source,

$$\mathfrak{o}_+ = \Pi_{\ker \mathrm{ad}_{h_0}} S, \tag{39}$$

the projection onto processes satisfying exact free-energy conservation.

14.2 A finite oscillator fails only at special resonances

For the cubic PU interaction at generic unequal frequencies, the positive metric can be corrected through the computed orders — the explicit generators exist and the transformed Hamiltonian is Hermitian. But at

$$\omega_1 = 3\omega_2, \tag{40}$$

one high-frequency quantum can convert into three low-frequency quanta without changing the total energy, and the second-order correction is obstructed by exactly that conversion operator, $a_1 a_2^{\dagger 3} - a_1^\dagger a_2^3$. At $2\omega_1 = 3\omega_2$ a third-order obstruction appears. Imagine two musical notes: usually they are unrelated, but if one note has exactly three times the frequency of the other, one vibration can split into three without changing the total energy — at that point the interaction is perfectly synchronized with the system.

The 3:1 obstruction is more than a failure of one formula. Sufficiently excited resonant multiplets develop complex-conjugate energies, $E_\pm = E_R \pm i\Gamma$, at second order. This is formally defined perturbative PT breaking: [16] constructs the antilinear symmetry explicitly, proves that it commutes with the interacting Hamiltonian, and certifies the complex pair *exactly* in the effective resonant multiplet (by rational algebra and Sturm counting, not only numerically). Where the complex pair persists, no positive-definite invariant metric can make the Hamiltonian equivalent to an ordinary self-adjoint one. Two levels of conclusion must not be merged: this complex-spectrum statement is established in the verified resonant oscillator setting; the field-theory result below is presently a perturbative metric obstruction, not yet a demonstrated complex spectrum.

14.3 The momentum continuum turns exceptional resonances into ordinary scattering

For an oscillator, the frequencies are fixed numbers and exact resonances require special ratios. For a field, $E_a(k) = \sqrt{k^2 + m_a^2}$, and momenta vary continuously: one can satisfy

$$E_H(k_1) + E_L(k_2) = E_L(k_3) + E_L(k_4) \tag{41}$$

together with momentum conservation on open sets of scattering data, for *any* split masses. The key process is the branch-changing scattering $H + L \to L + L$. Paper V shows that the second-order positive-metric source is nonzero on an open scattering-shell subset (exactly, at a rational kinematic point, hence on an open neighborhood — not claimed at every kinematic point): the continuum turns the oscillator's isolated resonance obstruction into a generic scattering-shell obstruction. In a small mechanical system, resonance requires the machine to be tuned to a special ratio; in a field, particles can change their momentum, and even if their masses are not specially tuned, their motion can supply the exact energy match.

The canonical positive completion therefore remains meaningful locally and perturbatively away from resonant processes, but it cannot generally be deformed into a positive pointed-sector metric once branch-changing scattering becomes exactly energy conserving. One qualification is essential: this establishes that *no analytic deformation of the canonical metric* exists there — not that no conceivable positive completion of any kind exists. For the oscillator with certified complex energies the stronger conclusion is available; for the field theory the perturbative statement is what is proved.

14.4 Why the obstruction waits until second order

The perfect-square cubic vertex is exactly the Källén polynomial, $\mathcal{V}_3 = \frac{1}{2}\lambda_{\mathrm{K}}(p_1^2, p_2^2, p_3^2)$ (Section 13 notation). It vanishes for three ordinary massless legs and has a second-order zero at the confluent massless point; and the transported reality factors eliminate the first-order positive obstruction on the physically open decay channel. So the positive metric survives the first test. But at second order, two cubic interactions combine into a branch-changing $2 \to 2$ process, and the obstruction appears: a single interaction event is too constrained to cause the problem, while two interaction events in sequence can convert the particles while preserving total energy — and that is where the positive geometry fails.

14.5 A different structure survives: two mirror Jordan sectors

The perfect-square theory admits an exact rewriting, introduced by Bateman and Turok [7], who also identified its exchange symmetry as ghost parity and as charge conjugation for an internal $O(1,1)$ symmetry; the sector-level reading given here is what the interaction paper adds. Introducing an auxiliary field and the nonlinear change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$,

$$\mathcal{L} = -\partial_\mu U\, \partial^\mu V + \frac{\lambda^2}{2} U^2 V^2. \tag{42}$$

The constant zero-potential stationary backgrounds form two branches, $(U_0, 0)$ and $(0, V_0)$, and the exact exchange $U \leftrightarrow V$ maps one branch to the other. Around $(1, 0)$ the linearized equations are $\Box v = 0$, $\Box u = \lambda^2 v$ — a Jordan chain generated by the interaction itself; around $(0, 1)$ the chain has the opposite orientation. The complete theory thus has two mirror valleys: choosing one valley gives one orientation of the Jordan system, and the exact symmetry does not rearrange particles inside that valley — it carries the entire valley to its mirror.

Three distinctions matter here. The exact symmetry belongs globally to the *extended* (U, V) theory; its expression in the original (ϕ, ψ) variables is a chart-transition map, singular at the chosen perturbative vacuum; and it is not a ghost-number sign inside one Fock sector. It may ultimately form part of an antiunitary CPT-like structure, but what is proved is a linear sector exchange, not a quantum CPT theorem.

Finally, the ordinary split-theory ghost parity becomes singular at the merge. In the confluent Jordan basis $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$, the branch parity becomes

$$P_\delta = \begin{pmatrix} 0 & 2/\delta \\ \delta/2 & 0 \end{pmatrix}, \qquad P_\delta^2 = 1, \tag{43}$$

with no bounded limit on a single Jordan chain: as the two particle types merge, the old rule "one is even and one is odd" becomes infinitely badly conditioned. The only finite remnant is to exchange the entire Jordan system with its mirror copy: after doubling by the oppositely oriented sector, the singular parity becomes a finite sector exchange. This confluent result is the *linear bridge* to the independently exact nonlinear $U \leftrightarrow V$ symmetry: what the regulated parity converges to is the linearization of that exchange around the two mirror backgrounds — no stronger identification is claimed.

The relation between the two structures is now verified to be sharp: the interacting on-shell scattering matrix is *exactly* pseudo-Hermitian with respect to ghost-branch parity — the Krein structure of [7] survives the interaction untouched — while every positive metric fails, and the failing part is exactly the exchange-odd component. The two escapes from the ghost problem, positive metric and Krein pairing, therefore *separate* in the interacting theory: the second is not a hidden version of the first.

Terminology. *Positive pseudo-Hermitian metric*: a positive inner product obtained by changing the adjoint through a Dyson map. *Metric deformation*: an order-by-order correction of that inner product after an interaction is introduced. *Obstruction*: the energy-conserving part of the unwanted non-Hermitian interaction that cannot be generated by a metric correction. *Pointed sector*: a quantum construction built around one selected vacuum background. *Doubled theory*: a completion containing both mirror Jordan sectors. *Sector-exchange involution*: the exact map exchanging the two mirror backgrounds and reversing the Jordan-chain orientation. *Spectral breaking*: the appearance of complex-conjugate energy levels, which excludes a positive invariant metric where the complex spectrum persists. "Ghost parity" is used only with a qualifier: split branch-number parity, the exact $U \leftrightarrow V$ exchange, or a (future) antiunitary CPT-like operation are different objects.

15 Why gravity is not merely five copies of the oscillator

The scalar PU oscillator has no gauge symmetry. Gravity does.

Consider linearized scalar-free Einstein–Weyl gravity around flat spacetime: the Einstein term plus a tuned quadratic-curvature term ($\alpha R_{\mu\nu}^2 + \beta R^2$ with $\alpha = -3\beta$). The linearized spectrum contains a massless graviton with two helicities and a massive spin-2 branch with five helicities [11]. A naive polarization-by-polarization treatment might suggest that every massive helicity belongs to an independent PU pair. Gauge reduction shows otherwise [15]: the reduced phase space has the structure

$$\Gamma_{\text{phys}} \cong 2\,\Gamma_{\text{PU}} \;\oplus\; 2\,\Gamma_{-} \;\oplus\; \Gamma_{-}. \tag{44}$$

The two helicity-2 sectors are genuine fourth-order PU pairs: each contains a massless graviton mode and a massive ghost partner. The helicity-1 and helicity-0 sectors contain only *unpaired* second-order massive ghosts: their apparent massless partners lie entirely in the diffeomorphism gauge orbit and disappear under reduction.

$$\boxed{\text{Gauge reduction preserves PU pairing only in helicity } \pm 2.}$$

This is the first major way in which gravity differs from the scalar model.

16 Lorentz covariance forbids hybrid solutions

The five massive helicities $+2, +1, 0, -1, -2$ form one irreducible massive spin-2 representation. They are not five independent species that can be quantized separately: a Lorentz transformation can mix them.

By Schur's lemma, an invariant Hermitian form on the irreducible spin-2 polarization space is proportional to the identity. Its sign or reality assignment must therefore be uniform across all five helicities. This rules out a tempting hybrid construction in which the helicity-2 modes receive the positive PU metric while the helicity-1 and helicity-0 ghosts are treated in a Krein space: such an assignment is not Lorentz covariant.

The gravity paper [15] therefore finds exactly two covariant completion classes:

1. a positive pseudo-Hermitian completion in which the full massive spin-2 multiplet has uniformly rotated reality;
2. a Krein completion in which the gravitational field retains standard reality and the entire massive spin-2 multiplet carries a uniform negative sign relative to the positive massless graviton sector.

The two completions induce the same complex gauge-invariant covariance, while differing in their involution and completion. This is the gravitational version of the real-form distinction of Figure 1.

17 The conformal Jordan boundary

Let the coefficient of the Einstein term tend to zero while the quadratic-curvature coupling remains fixed. The massive spin-2 mass then tends to zero, and the different helicity sectors behave differently [15].

Helicity ± 2. The massless and massive branches merge, producing two $\Box^2$ Jordan systems. Each polarization contributes two configuration-space solutions, giving four modes in total.

Helicity ± 1. The vector ghosts have an ordinary massless limit. They remain second-order modes rather than becoming Jordan partners.

Helicity 0. The scalar kinetic normalization tends to zero. At the conformal point a new Weyl gauge symmetry appears, $\delta h_{\mu\nu} = 2\sigma\eta_{\mu\nu}$: the scalar direction becomes null in the presymplectic form and is removed by the quotient.

The conformal count is therefore $4 + 2 + 0 = 6$. The positive split-frequency metric runs to infinite Cartan distance and has no nondegenerate positive continuation through the Jordan boundary; the complex spectral covariance and the standard-reality Krein form can continue, provided the newly null Weyl direction is quotiented out. The algebra of observables changes rank at the boundary, so convergence must be formulated on a fixed regular family of observables — most naturally through gauge- and Weyl-invariant curvature quantities. (Boundary constructions that instead *remove* one branch of the solution space, as in [28], and the richer phase diagram at nonzero cosmological constant [29, 30], are related but distinct stories.)

What Paper V predicts for interacting gravity. The interaction results of Section 14 have not yet been carried out for gravity, but they supply the machinery and the expected shape. The massive spin-2 multiplet can participate in branch-changing processes such as $h + h \leftrightarrow M$, $h + M \leftrightarrow h + h$, and $M + M \leftrightarrow h + M$, depending on kinematics and the cubic vertices. The natural test is: construct the free positive real form of the massive spin-2 sector; transport the cubic Einstein–Weyl vertices; project their anti-Hermitian parts onto energy-conserving scattering shells; test whether the positive metric deformation is obstructed; and search for any doubled or CPT-compatible completion. Interacting gravity has *not* been shown to fail; what Paper V supplies is the obstruction machinery, and it makes branch-changing gravitational scattering the natural next calculation.

18 What was already solved, and what is new

It is important not to claim that earlier work failed to solve anything. Bender and Mannheim solved the free unequal-frequency oscillator's spectral and positive-norm problem by introducing a nontrivial metric [5, 6]. Bateman and Turok supplied a controlled Krein-space treatment of the massless resonant scalar theory, including a ghost-parity interpretation [7]. The new contribution is a classification and unification (Figure 3):

Paper I [**12**] reconstructs the Bender–Mannheim generator from the symplectic normal-form data and classifies the diagonalizers and the algebraic metric freedom.

Paper II [**13**] proves that the canonical Bender–Mannheim transformation is the minimum-Cartan-distortion representative, embeds the result in a general orthogonal projection principle, and separates the physical Dyson equivalence from the auxiliary identity-embedding obstruction.

Paper III [**14**] separates metric geometry, vacuum covariance, dynamical Jordan structure, and field representation theory; identifies the common spectral divided-difference covariance, its confluent limit, and its fourth-order Hadamard structure.

Paper IV [**15**] determines how gauge reduction and Lorentz covariance change the classification in gravity: the helicity stratification, the exclusion of covariant hybrid completions, and the conformal Jordan boundary.

Paper V [**16**] tests which free completions survive interactions: the deformation and obstruction theory of Section 14 — generic deformability, resonance obstructions, spectral complexification, the generic continuum shell obstruction, and the exact doubled sector-exchange symmetry.

In one paragraph: the five papers develop a progression from free classification to interacting obstruction theory. The first two derive and geometrically characterize the canonical positive Pais–Uhlenbeck metric. The third separates this metric from the universal complex vacuum covariance and follows the theory to its Jordan boundary. The fourth determines how gauge reduction and Lorentz covariance modify the classification in quadratic gravity. The fifth asks whether the free completions survive interactions: the positive metric deforms away from resonances but is obstructed by exactly energy-conserving branch-changing processes — on isolated rational loci in the oscillator and on generic scattering shells in the field theory — while at the massless perfect-square boundary an exact doubled exchange survives between two oppositely oriented Jordan sectors. The resulting question is no longer simply whether a ghost can be assigned positive norm, but which real form and completion remain compatible with the actual interactions of the theory.

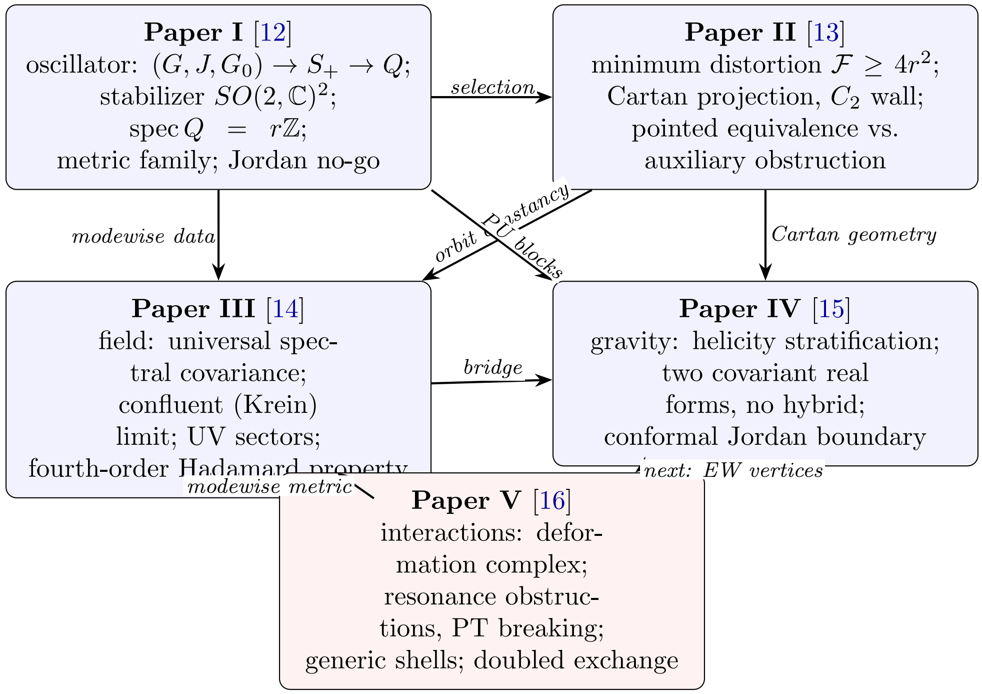

Figure 3: Dependency structure of the five technical papers.

The combined novelty is therefore not a new claim that one particular ghost Hamiltonian can be rewritten positively. It is the statement that:

> The known positive and Krein constructions are different real-form completions of a common complex spectral structure. Their admissibility and continuation are controlled by symplectic geometry, Cartan projection, gauge reduction, Lorentz representation theory, and Jordan degeneration.

19 What remains open

The first interaction-level answers are now in (Section 14), but they sharpen rather than exhaust the open questions.

For the positive completion: the deformation exists away from resonances, but does any nonanalytic or differently pointed positive completion evade the shell obstruction? Does the field theory, like the oscillator, develop complex energies (complex finite-volume levels or non-real resonance poles)? Does renormalization preserve the metric structure? (See [31] for a recent PT-based proposal at the interacting level.)

For the doubled structure: can the exact sector exchange be implemented as an operator on a quantum completion? Are the two pointed sectors superselected in infinite volume, or do parity vacua $(|0_A\rangle \pm |0_B\rangle)/\sqrt{2}$ exist? Can a probability interpretation be built on the doubled space, and is the exchange part of an antiunitary CPT-like operation?

For gravity: do the cubic Einstein–Weyl vertices obstruct the positive real form of the massive spin-2 sector through branch-changing scattering, as Paper V's machinery predicts is testable? Does the real-form classification survive on curved backgrounds and at nonzero cosmological constant [29, 30]?

These are no longer vague versions of "does the ghost go away?" They are concrete algebraic and analytic tests.

20 Conclusion

The traditional ghost problem combines several logically separate issues: classical stability, spectral boundedness, norm positivity, field reality, vacuum covariance, representation theory, gauge reduction, and Jordan degeneration.

The results now form a five-level hierarchy:

1. The free complex spectral theory can be identified.
2. Its positive and Krein real forms can be classified.
3. The canonical positive metric is geometrically selected.
4. Interactions test the deformability of these real forms.
5. On-shell conversion processes generate genuine obstructions.

The PU oscillator shows that a Hamiltonian that appears ghostlike in one real form may possess a positive pseudo-Hermitian completion in another; the minimum-distortion theorem explains why the Bender–Mannheim metric is geometrically canonical; the free field theory shows that the metric may become singular while the complex vacuum covariance remains regular; and gravity adds gauge symmetry and Lorentz covariance, which remove some pairings and force the massive spin-2 multiplet to be treated as a whole.

The interaction results complete the picture: the positive pseudo-Hermitian construction is a valid local resolution of the free and nonresonant weakly interacting theory. It is not generically stable under exactly energy-conserving branch-changing processes. In field theory, the momentum continuum makes such processes far more common than in a finite oscillator. At the massless perfect-square boundary, the surviving structure is instead an exact exchange between two oppositely oriented Jordan sectors.

The resulting view is neither that higher-derivative ghosts are automatically harmless nor that they automatically invalidate a theory. The refined question is:

Not "Can the ghost be removed?" but "Which completion survives the allowed interactions?"

Once the ingredients — complex dynamics, real form, involution, observable algebra, completion, and now interaction stability — are separated, the existing positive and indefinite constructions cease to look like unrelated tricks. They become different parts of one geometric, representation-theoretic, and deformation-theoretic framework.

Verification. Every finite-dimensional, variational, symplectic, and modewise distributional identity quoted in this introduction is machine-verified in the repository accompanying the series [12, 13, 14, 15, 16] (symbolic SymPy audits, high-precision numerical regression, and a Lean 4 formalization of the paper-I core with zero unproved placeholders); the technical papers state precisely which of their conclusions are analytic rather than machine-checked.

References

[1] M. Ostrogradsky, *Mémoire sur les équations différentielles relatives au problème des isopérimètres*, Mém. Acad. St. Pétersbourg **VI 4** (1850) 385.

[2] R. P. Woodard, *The theorem of Ostrogradsky*, arXiv:1506.02210.

[3] A. Pais and G. E. Uhlenbeck, *On field theories with non-localized action*, Phys. Rev. **79** (1950) 145.

[4] C. M. Bender and S. Boettcher, *Real spectra in non-Hermitian Hamiltonians having PT symmetry*, Phys. Rev. Lett. **80** (1998) 5243.

[5] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100** (2008) 110402.

[6] C. M. Bender and P. D. Mannheim, *Exactly solvable PT-symmetric Hamiltonian having no Hermitian counterpart*, Phys. Rev. D **78** (2008) 025022, arXiv:0804.4190.

[7] S. Bateman and N. Turok, *Escape from Ostrogradsky via hidden ghost parity*, arXiv:2607.00096.

[8] F. G. Scholtz, H. B. Geyer and F. J. W. Hahne, *Quasi-Hermitian operators in quantum mechanics and the variational principle*, Ann. Phys. **213** (1992) 74.

[9] A. Mostafazadeh, *Pseudo-Hermitian representation of quantum mechanics*, Int. J. Geom. Meth. Mod. Phys. **7** (2010) 1191.

[10] P. D. Mannheim, *Solution to the ghost problem in higher-derivative gravity*, arXiv:2109.12743.

[11] K. S. Stelle, *Classical gravity with higher derivatives*, Gen. Rel. Grav. **9** (1978) 353; Phys. Rev. D **16** (1977) 953.

[12] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper (Paper I) and verification repository, 2026.

[13] *The Pais–Uhlenbeck metric as a minimum-distortion principle, and the representation problem for the fourth-order field*, companion paper (Paper II), 2026.

[14] *The universal vacuum of the fourth-order scalar field: metric orbits, Fock sectors, and the Krein boundary*, companion paper (Paper III), 2026.

[15] *Gauge reduction and the completion problem in fourth-order gravity: PU pairing, covariant real forms, and the conformal Jordan boundary*, companion paper (Paper IV), 2026.

[16] *Interaction obstructions, resonant PT breaking, and doubled Jordan symmetry in fourth-order theories*, companion paper (Paper V), 2026.

[17] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.

[18] R. Bhatia, *Positive Definite Matrices*, Princeton University Press, 2007.

[19] G. D. Mostow, *Self-adjoint groups*, Ann. of Math. **62** (1955) 44.

[20] G. Kempf and L. Ness, *The length of vectors in representation spaces*, in: Algebraic Geometry, Lecture Notes in Math. **732**, Springer, 1979, 233.

[21] I. Gohberg, P. Lancaster and L. Rodman, *Indefinite Linear Algebra and Applications*, Birkhäuser, 2005.

[22] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1966.

[23] W. D. Heiss, *The physics of exceptional points*, J. Phys. A **45** (2012) 444016.

[24] J. von Neumann, *On infinite direct products*, Compos. Math. **6** (1939) 1.

[25] M. J. Radzikowski, *Micro-local approach to the Hadamard condition in quantum field theory on curved space-time*, Commun. Math. Phys. **179** (1996) 529.

[26] R. Brunetti and K. Fredenhagen, *Microlocal analysis and interacting quantum field theories*, Commun. Math. Phys. **208** (2000) 623.

[27] S. Hollands and R. M. Wald, *Local Wick polynomials and time ordered products of quantum fields in curved spacetime*, Commun. Math. Phys. **223** (2001) 289.

[28] J. Maldacena, *Einstein gravity from conformal gravity*, arXiv:1105.5632.

[29] H. Lü and C. N. Pope, *Critical gravity in four dimensions*, Phys. Rev. Lett. **106** (2011) 181302, arXiv:1101.1971.

[30] S. Deser, E. Joung and A. Waldron, *Partial masslessness and conformal gravity*, J. Phys. A **46** (2013) 214019, arXiv:1208.1307.

[31] J. Kubo and J. Kuntz, *Unitarity through PT symmetry in quantum quadratic gravity*, arXiv:2410.08278.